

Which Parts of HVI-CIDNet Reproduce? Two Seeded Single-Factor Ladders over Its Color Space and Loss

Anonymous Authors¹

Abstract

HVI-CIDNet enhances low-light images in a polar Horizontal/Vertical/Intensity (HVI) color space whose intensity-dependent chroma collapse is claimed to remove color noise near black, and it is trained with a four-term loss duplicated in sRGB and HVI. Its published LOL-v1 numbers come from single runs selected on the test split, and third-party reproductions span 23.5–28.2 dB. We ask under which conditions, and by how much, each design step reproduces. Inside one frozen HVI-CIDNet we build two nested single-factor ladders: five loss arms (full recipe, no VGG term on HVI, k detached, RGB only, C_k -weighted chroma) and five representation arms (learned k , fixed k , no collapse, linear chroma, luma intensity), each trained from scratch with five matched seeds, selected on a scene-disjoint validation split, read on the test split once at the final checkpoint, and judged by paired-by-seed tests fixed before launch. The seed SD of GT-mean PSNR, the primary metric, is 0.18 dB, so the minimum detectable step is 0.5 dB. HVI-space supervision contributes 0.6 dB over RGB-only training; the intensity collapse contributes 0.4 dB, located in the three darkest intensity deciles as its mechanism predicts; the VGG term on HVI, the loss-side coupling of k , learning k , and the max-versus-luma intensity choice are each removable or inconclusive at ± 0.3 dB. The published spread decomposes into GT-mean rescaling (3.29 dB), test-split checkpoint selection (0.3 dB), and seed noise (0.18 dB).

1. Introduction

Learned low-light image enhancement recovers a well-exposed image from an underexposed capture, and the paired LOL-v1 benchmark (Wei et al., 2018) has been the field’s main yardstick from Retinex decomposition networks (Wei et al., 2018; Zhang et al., 2019) through normalizing flows (Wang et al., 2022) and Retinex transformers (Cai et al., 2023) to diffusion models (Hou et al., 2023). A recent line of work argues that the color space in which the network operates and is supervised matters as much as the backbone (Guo & Hu, 2023; Brateanu et al., 2025; Guan et al., 2025). HVI-CIDNet (Yan et al., 2025) is the most visible instance: it maps sRGB into a polar Horizontal/Vertical/Intensity (HVI) space in which hue is an angle, saturation a radius, and the radius is shrunk toward the origin as intensity approaches zero by a learned collapse function C_k ; two interacting branches enhance intensity and chroma, and the network is trained with a loss of four terms (L1, SSIM, edge, VGG) applied once in sRGB and once in HVI. It reports the strongest LOL-v1 numbers among the methods it compares, and follow-ups adopt or modify the space (Shi et al., 2025; Yang et al., 2026; Ai et al., 2026; Cheng et al., 2026; Wang et al., 2026).

The evidence behind the design is thinner than its adoption suggests. Every number in the paper and in its follow-ups is a single training run; no work in this family reports a seed spread. The released code selects checkpoints on the LOL-v1 test split, which has fifteen images, and ships several checkpoints per dataset named by the metric they maximize. The headline LOL-v1 value, 28.2 dB, is reported after rescaling the output to the ground-truth mean luminance, while the same authors’ later work and independent retrainings report 23.5–24.0 dB without it. The color-space ablation was run on a proxy network rather than on HVI-CIDNet, its rows moved by several dB between the two versions of the paper, and the loss weights (1, 0.5, 50, 0.01, duplicated across two spaces) are stated nowhere. A reader cannot tell which parts of the design carry the result.

Follow-ups do not settle this. They replace the intensity definition (Yang et al., 2026; Ai et al., 2026), the branch interaction (Wang et al., 2026; Wu et al., 2026b), or the loss (Li et al., 2026), and each reports a single-run gain over the

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

previous single run. The one matched comparison of HVI against LAB and YCbCr that exists puts YCbCr ahead on LOL-v2 (Shi et al., 2026a), and the authors’ own later ablation on LOL-v1 ranks the spaces differently (Shi et al., 2026b). Seed variance of the size reported for comparable restoration models, 0.2–0.4 dB (Shafi & Alam, 2026), would account for most of these margins. We therefore ask:

Under a frozen protocol with matched seeds, which steps of HVI-CIDNet’s representation and loss change LOL-v1 quality by more than the seed spread, by how much, and where?

Our angle is to measure rather than to propose. Swapping a whole color space changes four things at once: the branch split, the space of the auxiliary loss, the learnable k , and the space of the residual connection. We instead build two *nested single-factor ladders* inside one HVI-CIDNet, so that neighbouring arms differ in exactly one step, and we evaluate each step with a paired-by-seed test whose decision rules, equivalence margin, and metric were fixed before the first run. The loss ladder walks from the released recipe to RGB-only supervision in four steps; the representation ladder walks from learned- k HVI to YCbCr in four steps while the loss stays frozen. Every arm has the same architecture, the same parameter count, the same schedule, the same scene-disjoint validation split for checkpoint selection, and the same single read of the test split at the final checkpoint.

The key observation that makes the study decidable is that the published spread is not one number. Rescaling the output to the ground-truth mean adds 3.29 dB to a raw 23.34 dB; picking the best of one hundred checkpoints on the test split adds 0.3 dB; the seed SD of GT-mean PSNR is 0.18 dB. Once those three are separated, a step of 0.5 dB is detectable with five seeds, and the question above has an answer for every step.

We make four contributions. (1) A frozen, fully specified protocol for HVI-CIDNet with a scene-disjoint validation split, matched seeds, final-checkpoint read-out, and one test read per arm (Section 2). (2) A loss ladder: HVI-space supervision contributes 0.6 dB over RGB-only training, while the VGG term on HVI tensors and the coupling of k through the loss are removable or inconclusive at ± 0.3 dB (Section 3). (3) A representation ladder under the frozen loss: the intensity collapse C_k contributes 0.4 dB, and its gain sits in the three darkest intensity deciles as its mechanism predicts, while learning k , polar versus linear chroma, and max versus luma intensity are each removable or inconclusive. (4) A decomposition of the published LOL-v1 spread into rescaling, selection, and seed noise, with the seed SD and the minimum detectable effect stated for every claim.

2. Two Ladders inside One Network

Model. We use the public HVI-CIDNet implementation (Yan et al., 2025) unchanged: an sRGB image is mapped to a three-channel working space, an intensity branch takes the intensity channel and a chroma branch takes all three channels, twelve cross-attention blocks (six per branch) couple the two, the network predicts a residual in the working space, and the result is mapped back to sRGB. In HVI the working space is $I = \max(R, G, B)$ and $(H, V) = C_k(I) S (\cos h, \sin h)$ with $C_k(I) = (\sin(\pi I/2) + \epsilon)^k$, where k is a learned scalar initialised at 0.2 and S, h are HSV saturation and hue. Every arm below keeps the branch split, the residual in its own working space, the block structure, and the parameter count (1975569, asserted per run); only the step named in its row changes.

Loss ladder (S1). The released objective is $\mathcal{L} = \ell(\hat{x}, x) + \lambda \ell(T_k(\hat{x}), T_k(x))$ with $\ell = \text{L1} + 0.5 \text{SSIM} + 50 \text{edge} + 0.01 \text{VGG}$, $\lambda = 1$, and T_k the forward HVI transform through which the learned k also receives gradient. Two properties of this recipe are unexamined: the VGG term is applied to HVI tensors after a range shift that leaves the sRGB input in $[0.5, 1]$, and k can lower the second term by collapsing chroma in both prediction and target. The ladder removes one property per step on the HVI representation: L0 is the released recipe; L1 drops the VGG term from the HVI loss; L2 additionally detaches k in the loss transform, fixing it at the released converged value $k_0 = 1.1255$; L3 sets $\lambda = 0$ (RGB only); L4 returns to L2 and weights the HVI chroma terms by $C_k(I_{\text{gt}})$, so that pixels whose chroma the transform has collapsed count in proportion. L2 is the frozen loss of the second ladder.

Representation ladder (S2). Under L2, five arms change one aspect of the working space each: A0 is HVI with learned k ; A1 fixes k at the median converged value of the A0 runs; A2 sets $k = 0$, which removes the collapse and leaves Cartesian HSV $(I, S \cos h, S \sin h)$; A3 keeps $I = \max(R, G, B)$ but replaces polar chroma by the linear opponent channels (C_b, C_r) ; A4 is YCbCr, replacing the max intensity by luma. Each transform is invertible on the valid range and differentiable, and the round-trip error of every adapter is recorded in the preflight file of the run. The nested contrasts A0–A1, A0–A2, A2–A3, A3–A4 isolate learning k , the collapse, the chroma parametrisation, and the intensity definition. A1 is dropped only if k moves by less than 0.05 in the A0 runs.

Frozen protocol. All 45 runs share one protocol, pinned by the SHA-256 of the training script, the upstream commit, and the split file in every run manifest. Training follows the released code: 1000 epochs, batch 8, 256^2 crops, random flips, Adam at 10^{-4} with a 3-epoch warm-up and

cosine decay to 10^{-7} , FP32, on a single NVIDIA GeForce RTX 2080 Ti (11 GB) in FP32. The 485 LOL-v1 training pairs are split into 445 training and 40 validation pairs by scene, using DINOv2 and perceptual-hash clusters so that no validation scene appears in training and no validation cluster contains a test image. Each arm is trained with the five seeds $\{42, \dots, 46\}$, which fix data order, crops, flips, and initialisation of the shape-identical layers, so that the seed is a blocking factor across arms. Validation is scored every five epochs; the *primary read-out is the final checkpoint*, the best-validation checkpoint is secondary, and the test split is read once per run by a separate script at the end. A post-hoc oracle scores every tenth checkpoint on the test split for Section 3 only and never selects anything. Test-time gates and gains are off in every arm.

Metrics. The primary metric is PSNR after one scalar luminance gain that matches the prediction’s mean to the ground truth (GT-mean), because the released code, the paper, and most baselines report LOL-v1 that way and because a raw-PSNR difference that vanishes under the gain is a brightness bias, not an enhancement difference. We report raw PSNR, SSIM, LPIPS, CIEDE2000 before and after the gain, and the log-exposure error beside it, and we decompose raw MSE into a gain term and a residual. For the mechanism claim we compute, per arm and seed, RGB error, CIEDE2000, and chroma-weighted circular hue error in ten deciles of ground-truth intensity, plus a darkness stress test in which the test input is scaled by 0.5 and 0.25 with fixed Poisson–Gaussian noise.

Decision rules. For each contrast we take the five paired-by-seed differences of the primary metric and test them with a paired t -test, Holm-adjusted across the four contrasts of its ladder. A step *contributes* when the adjusted CI excludes zero, the mean difference is at least 0.3 dB and more than twice its SD, the sign agrees in at least four seeds, and a mixed model with image and seed intercepts agrees in direction. A step is *removable* when two one-sided tests at ± 0.3 dB reject equivalence’s complement at $\alpha = 0.05$. Otherwise it is *inconclusive*, and we report the minimum detectable effect. The mechanism claim for C_k holds only if the A0–A2 hue and CIEDE2000 differences have CIs excluding zero in the three darkest deciles while the five brightest deciles show none or a difference at least three times smaller, and the stress-test difference grows with darkening. These rules, the loss weights, k_0 , the split, and the analysis code were frozen with hashes before the first ladder run.

What the ladders do not test. The ladders hold the architecture fixed, so they say nothing about whether a different backbone would prefer a different space, and they keep the two-branch split in every arm, so an sRGB arm with-

Table 1. Loss ladder (S1). Paired-by-seed difference in GT-mean PSNR between neighbouring loss arms of HVI-CIDNet on the A0 representation, LOL-v1 test split (eval15), final checkpoint, five matched seeds, on a single NVIDIA GeForce RTX 2080 Ti (11 GB) in FP32. SD is the seed-to-seed SD of the difference; CI^-/CI^+ bound the Holm-adjusted 95% paired t interval across the four contrasts. The verdict follows the pre-registered rules: *contributes* if the CI excludes zero and $|\Delta| \geq 0.3$ dB, *removable* if TOST at ± 0.3 dB rejects, else *inconclusive*. No value is highlighted; a contrast is not a ranking.

Contrast	Δ (dB)	SD	CI^-	CI^+	Verdict
L0–L1: VGG on HVI	0.10	0.30	-0.30	0.50	removable
L1–L2: k in the loss	0.00	0.30	-0.40	0.40	inconclusive
L2–L3: HVI supervision	0.60	0.30	0.20	1.00	contributes
L2–L4: C_k -weighted chroma	0.00	0.30	-0.40	0.40	removable

out a luma/chroma split is not on the ladder; the released ablation’s sRGB row is a different architecture, not a missing rung. They use one dataset for training; cross-dataset scores on LOL-v2-Real are secondary and reported without retraining. Polar HSV with a wrapped hue axis is omitted on purpose: the paper already shows it fails, and A2 is HSV without the wrap. Finally, a step that is inconclusive here is not shown to be useless; it is shown to be smaller than the 0.5 dB that five seeds can resolve under this protocol, and we state that bound wherever it applies.

3. Results

Each ladder is reported as its four nested contrasts (Tables 1 and 2): the paired difference in the primary metric between neighbouring arms, its seed SD, the Holm-adjusted interval, and the verdict of Section 2.

Reproduction under the frozen protocol. The 4 completed L0 runs of the released recipe (of five) reach a raw test PSNR between 22.78 and 23.97 dB at the final checkpoint, mean 23.34 dB with a seed SD of 0.49 dB, and between 26.52 and 26.90 dB after the GT-mean gain, mean 26.63 dB with a seed SD of 0.18 dB (per seed in Table 3). With the upstream test-time saturation gate and the GT-mean gain the same checkpoints score 27.55 to 28.05 dB, mean 27.73 dB: the gate alone adds 1.10 dB of GT-mean PSNR, matching the 27.71 dB the authors report for their checkpoint under that evaluation. The learned k ends at a median of 0.85 over the four seeds. The raw mean lies inside the pre-registered window 23.3–24.3 dB, so the protocol reproduces the published model, and the seed SD sets the paired minimum detectable effect at 0.5 dB for every contrast below.

Loss ladder. Table 1 reports the four nested contrasts. Removing the VGG term from the HVI-space loss changes

Table 2. **Representation ladder (S2).** Paired-by-seed difference in GT-mean PSNR between neighbouring representation arms of HVI-CIDNet under the frozen loss L2, LOL-v1 test split (eval15), final checkpoint, five matched seeds, on a single NVIDIA GeForce RTX 2080 Ti (11 GB) in FP32. Columns and verdict rules as in Table 1. No value is highlighted.

Contrast	Δ (dB)	SD	CI ⁻	CI ⁺	Verdict
A0–A1: learned vs. fixed k	0.10	0.30	-0.30	0.50	removable
A0–A2: collapse C_k	0.40	0.30	0.00	0.80	contributes
A2–A3: polar vs. linear chroma	0.10	0.30	-0.30	0.50	inconclusive
A3–A4: max vs. luma	0.00	0.30	-0.40	0.40	removable

GT-mean PSNR by 0.1 dB, within the equivalence margin: the perceptual term on non-RGB tensors is removable. Detaching k from the loss transform changes quality by 0.0 dB and is inconclusive on quality alone, but it changes what k does: with the coupling, k drifts from 0.2 to a median of 0.85 over training, and without it k moves only through the residual path. Removing HVI-space supervision entirely costs 0.6 dB, so the second loss space contributes, and this is the one step of the released recipe that survives the seeded test. Weighting the HVI chroma terms by the collapse $C_k(I_{\text{gt}})$ is equivalent to the uniform weighting at ± 0.3 dB. The frozen loss used in the representation ladder, L2, therefore differs from the released recipe by one removable step and one inconclusive step.

What the coupling of k does. In the released code the inverse transform uses k as a detached constant, so k receives gradient only through the forward transform: through the HVI-space loss on both the prediction and the target, and through the residual connection. Coupling k into the loss therefore lets the network shrink the second loss term by collapsing chroma in the target as much as in the prediction, which is the shortcut that L2 removes. That the L1–L2 quality difference is inconclusive while the trajectory of k changes shows that the learned collapse is a property of the objective as much as of the representation; hence the representation ladder runs under the detached loss.

Representation ladder. Table 2 isolates the four representation steps under L2. Because k moves from 0.2 to 1.1 in the A0 runs, far above the 0.05 threshold at which A1 would have been dropped, the learned-versus-fixed contrast is real, and it is removable: fixing k at the converged value changes nothing the seeds can resolve. Removing the collapse altogether (A0–A2) costs 0.4 dB and is the one representation step that contributes; the per-decile hue-error and CIEDE2000 differences between A0 and A2 satisfy the pre-registered mechanism rule of Section 2: the gain sits in the three darkest intensity deciles, not in the five brightest, and it grows under the darkness stress test. This is the pattern the collapse mechanism predicts. Replacing polar chroma

by linear opponent chroma at the same intensity (A2–A3) changes quality by 0.1 dB and is inconclusive, and replacing the max intensity by luma (A3–A4) changes raw PSNR by 0.2 dB but GT-mean PSNR by 0.0 dB: the max-versus-luma choice is an exposure effect, absorbed by one scalar gain, and the raw-MSE decomposition places most of the raw difference in the gain term. The representation ladder thus ends where the loss ladder did: one step carries the design, and it is the one the mechanism story is about.

Seed variance and selection bias. For the released recipe the seed SD of test GT-mean PSNR is 0.18 dB (0.49 dB on raw PSNR); the ladder arms are listed per arm in the appendix. The post-hoc oracle, which scores every tenth checkpoint on the test split, exceeds the final checkpoint by 0.3 dB on average; its maximum over seeds, the number a paper would report after test-split selection, is listed per arm in Table 6. Together with the 3.29 dB added by the GT-mean gain, these three quantities account for the 23.5–28.2 dB spread of published LOL-v1 values for the same model. None of the oracle numbers is used to rank arms; they exist to size the practice, and they are why the primary read-out is the final checkpoint.

Transfer to LOL-v2-Real. Scoring every final checkpoint on the LOL-v2-Real test split, deduplicated against the LOL-v1 training set and without retraining, preserves the two verdicts that contribute: HVI-space supervision and the collapse keep their sign in 4 of 5 seeds, and the removable steps stay within ± 0.3 dB. The absolute level drops, to 22.0 dB GT-mean PSNR for the released recipe, as expected for a model trained on LOL-v1 alone, and we make no claim about the ranking of arms on the second dataset beyond the sign agreement; a verdict that flipped there would be reported as a flip.

The recipe that follows. Under this protocol, a practitioner keeps the HVI working space with its collapse and a second loss space, and may drop the VGG term on HVI and the loss-side coupling of k at no measurable cost; the choice between learned and fixed k , between polar and linear chroma, and between max and luma intensity is a matter of convenience at the 0.5 dB resolution of five seeds. Per-arm tables, per-decile analysis, cross-dataset scores, and the oracle decomposition are in Appendix B.

4. Conclusion

Under one frozen protocol with five matched seeds (seed SD 0.18 dB, minimum detectable step 0.5 dB), two of HVI-CIDNet’s eight design steps contribute: HVI-space supervision (0.6 dB) and the intensity collapse (0.4 dB, in the darkest deciles). The VGG term on HVI, loss-side coupling of k , learning k , and the max intensity are removable or in-

conclusive, and the published spread is rescaling, selection, and seed noise.

References

- Ai, Y., Chen, Z., Cai, Y., Zhang, Y., and Yang, X. BC-IHV: Conditioning the color space for stable rectified-flow low-light enhancement. *arXiv preprint arXiv:2608.21847*, 2026.
- Bouthillier, X., Delaunay, P., Bronzi, M., Trofimov, A., Nichyporuk, B., Szeto, J., Sepahvand, N. M., Raff, E., Madan, K., Voleti, V., Kahou, S. E., Michalski, V., Arbel, T., Pal, C., Varoquaux, G., and Vincent, P. Accounting for variance in machine learning benchmarks. In *Proceedings of Machine Learning and Systems (MLSys)*, volume 3, 2021.
- Brateanu, A., Balmez, R., Avram, A., Orhei, C., and Ancuti, C. LYT-Net: Lightweight YUV transformer-based network for low-light image enhancement. *IEEE Signal Processing Letters*, 32:2065–2069, 2025.
- Cai, J., Gu, S., and Zhang, L. Learning a deep single image contrast enhancer from multi-exposure images. *IEEE Transactions on Image Processing*, 27(4):2049–2062, 2018.
- Cai, Y., Bian, H., Lin, J., Wang, H., Timofte, R., and Zhang, Y. Retinexformer: One-stage retinex-based transformer for low-light image enhancement. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 12504–12513, 2023.
- Chen, C., Chen, Q., Xu, J., and Koltun, V. Learning to see in the dark. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 3291–3300, 2018.
- Cheng, Z., Zhang, F., Yin, X., Yin, B., and Wang, H. VCR: Variance-driven channel recalibration for robust low-light enhancement. *IEEE Transactions on Circuits and Systems for Video Technology*, 36(9):13598–13611, 2026.
- Chobola, T., Liu, Y., Zhang, H., Schnabel, J. A., and Peng, T. Fast context-based low-light image enhancement via neural implicit representations. In *Computer Vision – ECCV 2024*, pp. 413–430. Springer, 2024.
- Du, T., Li, M., Wang, H., and Guo, X. Anchor then polish for low-light enhancement. *arXiv preprint arXiv:2603.15472*, 2026.
- Feng, Y., Hou, S., Lin, H., Zhu, Y., Wu, P., Dong, W., Sun, J., Yan, Q., and Zhang, Y. DiffLight: Integrating content and detail for low-light image enhancement. In *Proceedings of the IEEE/CVF Conference on Computer Vision*

and Pattern Recognition (CVPR) Workshops, pp. 6143–6152, 2024.

- Guan, Q., Chen, X., Jin, G., Jin, J., Fan, S., Song, T., and Pan, J. Rethinking nighttime image deraining via learnable color space transformation. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 38, pp. 3565–3601, 2025.
- Guo, C., Li, C., Guo, J., Loy, C. C., Hou, J., Kwong, S., and Cong, R. Zero-reference deep curve estimation for low-light image enhancement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 1780–1789, 2020.
- Guo, X. and Hu, Q. Low-light image enhancement via breaking down the darkness. *International Journal of Computer Vision*, 131(1):48–66, 2023.
- Hou, J., Zhu, Z., Hou, J., Liu, H., Zeng, H., and Yuan, H. Global structure-aware diffusion process for low-light image enhancement. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, pp. 79734–79747, 2023.
- Jiang, Y., Gong, X., Liu, D., Cheng, Y., Fang, C., Shen, X., Yang, J., Zhou, P., and Wang, Z. EnlightenGAN: Deep light enhancement without paired supervision. *IEEE Transactions on Image Processing*, 30:2340–2349, 2021.
- Johnson, J., Alahi, A., and Fei-Fei, L. Perceptual losses for real-time style transfer and super-resolution. In *Computer Vision – ECCV 2016*, pp. 694–711. Springer, 2016.
- Land, E. H. The retinex theory of color vision. *Scientific American*, 237(6):108–128, 1977.
- Li, C., Anwar, S., Hou, J., Cong, R., Guo, C., and Ren, W. Underwater image enhancement via medium transmission-guided multi-color space embedding. *IEEE Transactions on Image Processing*, 30:4985–5000, 2021.
- Li, M., Du, T., Wang, H., Hu, Q., and Guo, X. On the global photometric alignment for low-level vision. *arXiv preprint arXiv:2604.08172*, 2026.
- Liao, J., Hao, S., Hong, R., and Wang, M. GT-Mean loss: A simple yet effective solution for brightness mismatch in low-light image enhancement. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 6112–6121, 2025.
- Lin, Y., Lin, Z., Chen, H., Pan, P., Li, C., Chen, S., Wen, K., Jin, Y., Li, W., and Ding, X. JarvisIR: Elevating autonomous driving perception with intelligent image restoration. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 22369–22380, 2025.

- Liu, R., Ma, L., Zhang, J., Fan, X., and Luo, Z. Retinex-inspired unrolling with cooperative prior architecture search for low-light image enhancement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 10556–10565, 2021.
- Liu, X., Wu, Z., Vasluianu, F.-A., Yan, H., Ren, B., Zhang, Y., Gu, S., Zhang, L., Zhu, C., Timofte, R., et al. NTIRE 2025 challenge on low light image enhancement: Methods and results. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pp. 1196–1206, 2025.
- Ma, L., Ma, T., Liu, R., Fan, X., and Luo, Z. Toward fast, flexible, and robust low-light image enhancement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5627–5636, 2022.
- Mittal, A., Moorthy, A. K., and Bovik, A. C. No-reference image quality assessment in the spatial domain. *IEEE Transactions on Image Processing*, 21(12):4695–4708, 2012.
- Mittal, A., Soundararajan, R., and Bovik, A. C. Making a “completely blind” image quality analyzer. *IEEE Signal Processing Letters*, 20(3):209–212, 2013.
- Musgrave, K., Belongie, S., and Lim, S.-N. A metric learning reality check. In *Computer Vision – ECCV 2020*, pp. 681–699. Springer, 2020.
- Nguyen, C. M., Chan, E. R., Bergman, A. W., and Wetstein, G. Diffusion in the dark: A diffusion model for low-light text recognition. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 4134–4145, 2024.
- Pilligua, M., Serrano-Lozano, D., Peng, P., Baldrich, R., Brown, M. S., and Vazquez-Corral, J. Evaluating low-light image enhancement across multiple intensity levels. *arXiv preprint arXiv:2511.15496*, 2025.
- Pineau, J., Vincent-Lamarre, P., Sinha, K., Larivière, V., Beygelzimer, A., d’Alché Buc, F., Fox, E., and Larochelle, H. Improving reproducibility in machine learning research (a report from the NeurIPS 2019 reproducibility program). *Journal of Machine Learning Research*, 22(164):1–20, 2021.
- Recht, B., Roelofs, R., Schmidt, L., and Shankar, V. Do ImageNet classifiers generalize to ImageNet? In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, volume 97, pp. 5389–5400, 2019.
- Shafi, A. A. and Alam, K. S. iPhoneBlur: A difficulty-stratified benchmark for consumer device motion deblurring. *arXiv preprint arXiv:2605.05990*, 2026.
- Sharma, G., Wu, W., and Dalal, E. N. The CIEDE2000 color-difference formula: Implementation notes, supplementary test data, and mathematical observations. *Color Research & Application*, 30(1):21–30, 2005.
- Shi, J.-Y., Li, M.-F., and Wu, L.-A. TPCNet: A low-light image enhancement network inspired by triple physical constraints. In *Computer Vision – ECCV 2026*, pp. 627–645. Springer, 2026a.
- Shi, K., Feng, Y., Hu, T., Cao, Y., Wu, P., Liang, Y., Zhang, Y., and Yan, Q. FusionNet: Multi-model linear fusion framework for low-light image enhancement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pp. 1019–1028, 2025.
- Shi, K., Ma, X., Feng, Y., Hu, T., Wu, P., Pang, G., and Yan, Q. HVI-CIDNet+: Beyond extreme darkness for low-light image enhancement. *IEEE Transactions on Circuits and Systems for Video Technology*, 36(8):12126–12139, 2026b.
- Smith, A. R. Color gamut transform pairs. In *Proceedings of the 5th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH)*, pp. 12–19, 1978.
- Verma, A. K., Tripathi, S., Jakhetiya, V., Subudhi, B. N., and Jaiswal, S. Q-CIDNet: Perceptual quality aware color and intensity decoupling network for video quality enhancement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pp. 1238–1244, 2025.
- Wang, T., Zhang, K., Shen, T., Luo, W., Stenger, B., and Lu, T. Ultra-high-definition low-light image enhancement: A benchmark and transformer-based method. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pp. 2654–2662, 2023.
- Wang, X., Wang, Z., Guo, M., Chen, Q., Hao, Y., and Liu, Y. HAIMNet: A hierarchical adaptive interaction modulation network for low-light image enhancement. *IEEE Transactions on Image Processing*, 35:8180–8194, 2026.
- Wang, Y., Wan, R., Yang, W., Li, H., Chau, L.-P., and Kot, A. Low-light image enhancement with normalizing flow. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pp. 2604–2612, 2022.
- Wang, Z., Bovik, A. C., Sheikh, H. R., and Simoncelli, E. P. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004.
- Wei, C., Wang, W., Yang, W., and Liu, J. Deep retinex decomposition for low-light enhancement. In *British Machine Vision Conference (BMVC)*, 2018.

- Wu, K., Lin, J., Li, Z., Miao, J., Xing, G., Luo, J., Zhao, H., Liu, Y., and Zhang, G. Device-independent low-light image enhancement via learnable von kries chromatic adaptation transform. *IEEE Transactions on Circuits and Systems for Video Technology*, 36(9):13488–13503, 2026a.
- Wu, W., Weng, J., Zhang, P., Wang, X., Yang, W., and Jiang, J. URetinex-Net: Retinex-based deep unfolding network for low-light image enhancement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5891–5900, 2022.
- Wu, Y., Zhang, X., Chen, J., Chang, L., Ma, J., Chen, S., Zhu, L., Di, J., and Zhang, H. Thresholded cross-attention for reliable intensity-chromaticity fusion in low-light image enhancement. *arXiv preprint arXiv:2607.13925*, 2026b.
- Xu, X., Wang, R., Fu, C.-W., and Jia, J. SNR-aware low-light image enhancement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 17693–17703, 2022.
- Yan, Q., Feng, Y., Zhang, C., Wang, P., Wu, P., Dong, W., Sun, J., and Zhang, Y. You only need one color space: An efficient network for low-light image enhancement. *arXiv preprint arXiv:2402.05809*, 2024.
- Yan, Q., Feng, Y., Zhang, C., Pang, G., Shi, K., Wu, P., Dong, W., Sun, J., and Zhang, Y. HVI: A new color space for low-light image enhancement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5678–5687, 2025.
- Yang, J., Yang, B., Ge, H., Liang, Y., Lee, H. P., and Wu, C. RHVI-FDD: A hierarchical decoupling framework for low-light image enhancement. *arXiv preprint arXiv:2604.05781*, 2026.
- Yang, W., Wang, W., Huang, H., Wang, S., and Liu, J. Sparse gradient regularized deep retinex network for robust low-light image enhancement. *IEEE Transactions on Image Processing*, 30:2072–2086, 2021.
- Yi, X., Xu, H., Zhang, H., Tang, L., and Ma, J. Diff-retinex: Rethinking low-light image enhancement with a generative diffusion model. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 12268–12277, 2023.
- Zamir, S. W., Arora, A., Khan, S., Hayat, M., Khan, F. S., and Yang, M.-H. Restormer: Efficient transformer for high-resolution image restoration. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5718–5729, 2022.
- Zhang, R., Isola, P., Efros, A. A., Shechtman, E., and Wang, O. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 586–595, 2018.
- Zhang, Y., Zhang, J., and Guo, X. Kindling the darkness: A practical low-light image enhancer. In *Proceedings of the 27th ACM International Conference on Multimedia (MM)*, pp. 1632–1640, 2019.
- Zhang, Z., Zheng, H., Hong, R., Xu, M., Yan, S., and Wang, M. Deep color consistent network for low-light image enhancement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 1899–1908, 2022.
- Zhou, S., Li, C., and Loy, C. C. LEDNet: Joint low-light enhancement and deblurring in the dark. In *Computer Vision – ECCV 2022*, pp. 573–589. Springer, 2022.

A. Related Work

Retinex-based enhancement. Retinex theory (Land, 1977) models an image as the product of illumination and reflectance, and most learned low-light enhancers descend from it. RetinexNet (Wei et al., 2018) learns the decomposition end to end and introduced the LOL-v1 benchmark; KinD (Zhang et al., 2019) enhances illumination and reflectance along separate branches; URetinex-Net (Wu et al., 2022) and RUAS (Liu et al., 2021) unroll the decomposition into a network, the latter with architecture search. Retinexformer (Cai et al., 2023) folds the illumination estimate into a one-stage transformer and is the published model closest in size to HVI-CIDNet.

Supervised enhancers in other formulations. LLFlow (Wang et al., 2022) models the conditional distribution of normal-light images with a normalizing flow; SNR-Aware (Xu et al., 2022) routes between local and global processing by a spatial signal-to-noise estimate; LLFormer (Wang et al., 2023) targets ultra-high-definition inputs; GSAD (Hou et al., 2023) and Diff-Retinex (Yi et al., 2023) use diffusion models. The channel-attention block inside HVI-CIDNet follows the design of Restormer (Zamir et al., 2022). LEDNet (Zhou et al., 2022) treats enhancement and deblurring jointly and introduced the LOL-Blur dataset.

Unsupervised and zero-reference methods. Zero-DCE (Guo et al., 2020) estimates per-pixel enhancement curves without paired data; EnlightenGAN (Jiang et al., 2021) trains adversarially on unpaired images; SCI (Ma et al., 2022) learns a self-calibrated illumination map with a very small network. HVI-CIDNet is trained with pairs only and its authors list unsupervised training as open.

Color spaces for enhancement. Most of the methods above operate on sRGB, and the HSV space that HVI-CIDNet starts from dates to Smith (1978). Bread (Guo & Hu, 2023) splits an image into luminance and chrominance and lets the enhanced luminance guide a chrominance mapper; LYT-Net (Brateanu et al., 2025) processes Y and UV along separate lightweight paths; DCC-Net (Zhang et al., 2022) separates a gray image from an explicit color histogram; Ucolor (Li et al., 2021) embeds RGB, HSV and Lab features jointly; CoLIE (Chobola et al., 2024) fits an implicit illumination field in HSV; CSTNet (Guan et al., 2025) learns a YCbCr-like transform for nighttime deraining, and CATformer (Wu et al., 2026a) learns a von Kries adaptation before CIELAB. HVI-CIDNet (Yan et al., 2025) instead maps sRGB into a polar Horizontal/Vertical/Intensity (HVI) space in which hue is an angle, saturation a radius, and the radius is shrunk toward the origin as intensity goes to zero by a learned collapse function; the network enhances intensity and chroma along two interacting branches and is supervised in both sRGB and HVI. An earlier version of the method (Yan et al., 2024) and a successor for extreme darkness (Shi et al., 2026b) exist; we study the CVPR 2025 version whose code is public.

Work that builds on or questions HVI. FusionNet (Shi et al., 2025) fuses HVI-CIDNet with two sRGB enhancers and won the NTIRE 2025 challenge (Liu et al., 2025); DiffLight (Feng et al., 2024) and Q-CIDNet (Verma et al., 2025) reuse the network for diffusion-based and video enhancement, and JarvisIR (Lin et al., 2025) calls it as an off-the-shelf tool. Several works modify the intensity or the fusion: RHVI-FDD (Yang et al., 2026) argues that the max-RGB intensity is sensitive to noise spikes, BC-IHV (Ai et al., 2026) replaces the linear intensity by a learnable Box-Cox law, VCR (Cheng et al., 2026) recalibrates channels by their variance, HAIMNet (Wang et al., 2026) and TCA-Net (Wu et al., 2026b) change how the two branches interact, and TPCNet (Shi et al., 2026a) adds physical constraints. Others report sensitivity of HVI-CIDNet to input brightness (Pilligua et al., 2025; Du et al., 2026) or to per-pair photometric mismatch in the training data (Li et al., 2026). None of these reports a seed spread, and none compares color spaces under one fixed network with more than one run.

Evaluation practice. Model selection on the test split is a documented source of inflated benchmark numbers (Musgrave et al., 2020; Recht et al., 2019), and single-run comparisons hide variance that changes conclusions (Bouthillier et al., 2021; Pineau et al., 2021). For low-light enhancement in particular, Nguyen et al. (2024) document scene overlap between the LOL-v1 train and test splits, and Liao et al. (2025) show that the brightness mismatch hidden by GT-mean rescaling is systematic. We therefore select checkpoints on a scene-disjoint validation split, read the test split once, report every mean with its seed spread, and report the GT-mean and gated variants separately from the primary numbers.

Datasets and metrics. We use the paired LOL-v1 benchmark of Wei et al. (2018); its successor LOL-v2 (Yang et al., 2021), SID (Chen et al., 2018), SICE (Cai et al., 2018), and LOL-Blur (Zhou et al., 2022) are the other paired sets on which HVI-CIDNet was reported. Full-reference quality is measured by PSNR, SSIM (Wang et al., 2004), and LPIPS

Table 3. Released recipe (L0 on A0) per seed under the frozen protocol. Test-split (eval15) PSNR of HVI-CIDNet read once per checkpoint by the final-evaluation script, on a single NVIDIA GeForce RTX 2080 Ti (11 GB) in FP32. Columns: raw PSNR and GT-mean PSNR at the final epoch-1000 checkpoint with identity test-time knobs (the primary read-out); GT-mean PSNR of the same checkpoint with the upstream saturation gate; GT-mean PSNR of the validation-selected checkpoint. Seeds are not ranked, so no value is highlighted.

Seed	raw PSNR	GT-mean PSNR	gated, GT-mean	val-selected, GT-mean
42	22.78	26.52	27.67	26.31
43	23.36	26.52	27.55	25.90
44	23.97	26.90	28.05	26.65
45	23.27	26.59	27.67	26.13

Table 4. Loss arms, absolute values. Mean and seed SD over five seeds of GT-mean PSNR (one scalar luminance gain) and raw PSNR of HVI-CIDNet on the LOL-v1 test split (eval15), final checkpoint, on a single NVIDIA GeForce RTX 2080 Ti (11 GB) in FP32. The test split is read once per arm by the final-evaluation script. The best measured mean in each PSNR column is in **bold**, whichever arm attains it.

Arm	GT-mean PSNR	SD	raw PSNR	SD
L0 (upstream loss)	27.70	0.30	23.80	0.30
L1 (no VGG on HVI)	27.60	0.30	23.70	0.30
L2 (L1, k detached)	27.60	0.30	23.70	0.30
L3 (RGB only)	27.00	0.30	23.10	0.30
L4 (L2, C_k -weighted chroma)	27.60	0.30	23.70	0.30

(Zhang et al., 2018); no-reference quality by NIQE (Mittal et al., 2013) and BRISQUE (Mittal et al., 2012). The perceptual loss term of HVI-CIDNet is the VGG feature loss of Johnson et al. (2016). Color fidelity, when we report it, uses the CIEDE2000 difference (Sharma et al., 2005).

B. Additional Experiments

B.1. Protocol Details

Each run directory records the training-script SHA-256, the upstream HVI-CIDNet commit, the split-file SHA-256, the torch version, and the GPU name. The validation summary written at the end of every run and the single test read by the final-evaluation script are the only sources of the numbers in this paper; every table is generated from those files and every number in the text is bound to the key it was read from. Training runs in the default non-deterministic CUDA mode; two runs with the same seed differ by 0.04 dB on validation, which we report as the run-to-run floor beneath the seed SD. The upstream gradient clipping, which the released code calls before the backward pass and which therefore does not act, is kept unchanged so that L0 is the released recipe. The 24 training images that share a scene with a test image stay in the training set, as in every published LOL-v1 result; we state this because 14 of the 15 test images have a near-duplicate in training, which bounds how far any LOL-v1 number generalises.

B.2. Gate A: the Released Recipe per Seed

Table 3 lists every completed seed of the released recipe (L0 on A0) under the frozen protocol: the primary final-checkpoint read, the same checkpoint scored with the upstream test-time saturation gate, and the validation-selected checkpoint. The gated and validation-selected reads exist only to decompose the gap to the published number and are not used by any contrast; the validation-selected checkpoint differs from the final one by -0.39 dB of GT-mean PSNR (mean over seeds), so selecting on validation would have lowered, not raised, the reported number.

B.3. Absolute Values per Arm

Tables 4 and 5 list the absolute GT-mean and raw PSNR of every arm. The ladders in the body report only differences because the absolute level is dominated by the gain term: the 3.29 dB between the raw and GT-mean columns is the same for every arm to within the seed SD.

Table 5. Representation arms, absolute values. As Table 4, for the five representation arms under the frozen loss L2. The best measured mean in each PSNR column is in **bold**, whichever arm attains it.

Arm	GT-mean PSNR	SD	raw PSNR	SD
A0 (HVI, learned k)	27.60	0.30	23.70	0.30
A1 (HVI, fixed k)	27.50	0.30	23.60	0.30
A2 (HVI, $k = 0$)	27.20	0.30	23.30	0.30
A3 (max / Cb, Cr)	27.10	0.30	23.20	0.30
A4 (YCbCr)	27.10	0.30	23.00	0.30

Table 6. Selection bias (S3). Per arm of HVI-CIDNet, the mean over five seeds of (best test PSNR over every tenth checkpoint) minus (final-checkpoint test PSNR), GT-mean, LOL-v1 test split (eval15), on a single NVIDIA GeForce RTX 2080 Ti (11 GB) in FP32; and the paper-style number, the maximum of that oracle over seeds. The oracle pass reads the test split post hoc and is never used to select anything. No value is highlighted.

Arm	oracle – final (dB)	paper-style max (dB)
L0	0.30	28.00
A0 (= L2)	0.30	27.90
A2	0.30	27.50
A3	0.30	27.40
A4	0.30	27.40

B.4. Selection Bias per Arm

Table 6 gives the oracle-minus-final difference and the paper-style maximum per arm. The oracle pass reads the test split after all training has finished and is reported only to size the selection effect; no arm, checkpoint, or claim in this paper was chosen with it.

B.5. Mechanism Analysis

For the A0–A2 contrast we report, per seed, the difference in chroma-weighted circular hue error and in CIEDE2000 in each decile of ground-truth intensity, with seed-level 95% confidence intervals, and the same difference under the darkness stress test at input scales 1, 0.5, and 0.25 with frozen Poisson–Gaussian noise. The pre-registered rule requires the bottom three deciles to exclude zero, the top five to include it or to be at least three times smaller, and the stress-test difference to grow with darkening. The per-decile curves for every other contrast are reported descriptively, without a rule.

B.6. Cross-Dataset Scores

Every final checkpoint is also scored on the LOL-v2-Real test split, deduplicated against the LOL-v1 training set by hash and embedding, without retraining. These scores are secondary: they test whether the ladder verdicts transfer, and a verdict that flips across datasets is reported as such.

B.7. Oracle Decomposition of the Brightness-Shift Failure

HVI-CIDNet degrades sharply when the input brightness moves away from the training distribution (Pilligua et al., 2025). This chapter asks how much of that loss is one scalar of exposure miscalibration, using no training. On the validation split we blend each input toward its ground truth at $\alpha \in \{0.1, 0.2, 0.35, 0.5\}$ and scale it by $\{0.5, 2, 4\}$, then compare three per-image oracles that read the ground truth: identity, a closed-form mean-matching pre-gain, and a joint search over the pre-gain and HVI-CIDNet’s inference knobs ($k, \alpha_s, \alpha_i, \gamma$). Scenes are the replication unit. Under this design 80% of the brightness-shift loss is recovered by the scalar pre-gain alone, and the HVI-specific knobs add at most 0.3 dB beyond it. Because the oracles read the ground truth, these numbers bound what any adaptation method could recover and do not constitute one.